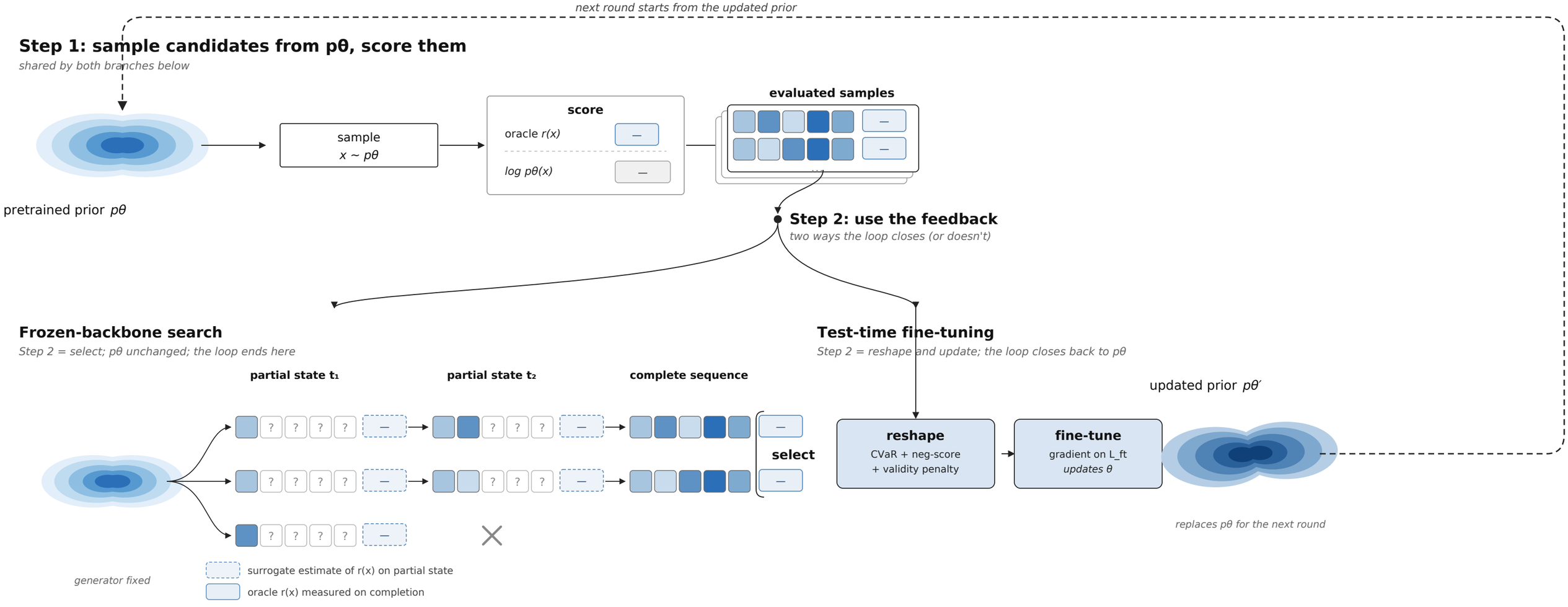
next round starts from the updated prior
Step 1: sample candidates from pθ, score them
shared by both branches below
pretrained prior pθ
sample
x ~ pθ
score
oracle r(x)
log pθ(x)
evaluated samples
Step 2: use the feedback
two ways the loop closes (or doesn't)
Frozen-backbone search
Step 2 = select; pθ unchanged; the loop ends here
partial state t₁
partial state t₂
complete sequence
select
generator fixed
surrogate estimate of r(x) on partial state
oracle r(x) measured on completion
Test-time fine-tuning
Step 2 = reshape and update; the loop closes back to pθ
reshape
CVaR + neg-score
+ validity penalty
fine-tune
gradient on L_ft
updates θ
updated prior pθ′
replaces pθ for the next round